DATA 211 – Introduction to Data Science
Topic 2b: NumPy and Matplotlib

Question bank: 40 multiple choice + 10 programming

Questions from DATA 211 / ISE 291 old exams and quizzes (AI_Work). **[new]** marks new questions written in the same style.

Part A: Multiple choice

Q1.

Topic 2 MCQ (T212)

What is the output of the following code?

```
np.zeros(3, dtype="int")
```

(A) `[[0, 0, 0], [0, 0, 0], [0, 0, 0]]`

(B) `array([0, 0, 0])`

(C) `array([0., 0., 0.])`

(D) `array([[0, 0, 0]])`

Answer: (B). A single integer gives a 1-D array of length 3; `dtype="int"` makes the zeros integers (no decimal point).

My answer

Q2.

Topic 2 practice questions

What is the output of the following code?

```
np.linspace(1, 2, 6)
```

(A) `array([1. , 1.2, 1.4, 1.6, 1.8, 2. ])`

(B) `array([1.0 , 1.1, 1.2, 1.4, 1.6, 2.0 ])`

(C) `list([1. , 1.2, 1.4, 1.6, 1.8, 2. ])`

(D) `([1. , 1.2, 1.4, 1.6, 1.8, 2. ])`

Answer: (A). `linspace` returns an array of 6 equally spaced values from 1 to 2 inclusive (step 0.2).

My answer

Q3.

Mid exam (important questions)

Which of the following statements most accurately describes *shape* in the context of NumPy arrays?

(A) How elements of an array are organized in rows and columns

(B) The total number of elements in an array

(C) Size of an array along each of its dimensions

(D) The number of dimensions of an array

Answer: (C). `shape` is a tuple giving the length along each axis; `size` is the total count and `ndim` the number of dimensions.

My answer

Q4.

Quiz 1 key (Version B)

What is the output?

```
import numpy as np
A = np.array([[2, 4],
              [6, 8]])
print(A.shape, A.ndim, A.size)
```

(A) `(4,), 1, 4`

(B) `(2, 2), 1, 2`

(C) `(2,), 2, 2`

(D) `(2, 2), 2, 4`

Answer: (D). 2 rows and 2 columns, 2 dimensions, 4 elements in total.

My answer

Q5.

Topic 2 practice questions

What is the output of the following code?

```
import numpy as np
np.random.seed(0)
x2 = np.random.randint(10, size=(3,4))
print(f"{x2.ndim}, {x2.shape},{x2.size}")
```

(A) `2, (3, 4),12`

(B) `2, (4, 5),20`

(C) `4, (4, 5),20`

(D) `2, (4, 5),9`

Answer: (A). `size=(3,4)` gives a 2-D array with shape (3, 4) and $3 \times 4 = 12$ elements.

My answer

Q6.

Quiz 1 key (T251)

Which of the following is NOT True about 2D NumPy arrays?

(A) arrays are flexible like Python lists (contain elements of different data types)

(B) arrays are mutable, like Python lists.

(C) arrays can be sliced along the rows or columns.

(D) A constant value can be added to any 2D array.

Answer: (A). A NumPy array holds elements of a single data type (one `dtype`); the other statements are true.

My answer

Q7.

Quiz 1 Sec 04 solution (T251)

What is the output of the following code?

```
import numpy as np
x2 = np.array([[7, 6, 1], [2, 5, 8], [3, 4, 9]])
print(x2[0,0], x2[1,2], x2[2,-1], x2[-2,-3])
```

(A) `7 8 2 9`

(B) `8 7 9 2`

(C) `7 8 9 2`

(D) `7 9 8 2`

Answer: (C). `x2[0,0]`=7, `x2[1,2]`=8, `x2[2,-1]`=9, `x2[-2,-3]` is row 1, column 0 = 2.

My answer

Q8.

Quiz 1 key (Version B)

What is the output?

```
import numpy as np
M = np.arange(1, 10).reshape(3, 3)
print(M[:2, 1:])
```

(A) `[[1 2] [4 5]]`

(B) `[[2 3] [5 6]]`

(C) `[[3 6] [2 5]]`

(D) `[[2 3 4] [5 6 7]]`

Answer: (B). M is [[1 2 3] [4 5 6] [7 8 9]]; rows 0–1 and columns 1 to the end give [[2 3] [5 6]].

My answer

Q9. Quiz 1 with solution (T252)

Given:

```
import numpy as np
a = np.arange(12).reshape(3,4)
b = a[:, ::-1]
print(b[1,2])
```

What will be printed?

(A) 6

(B) 7

(C) 5

(D) 9

Answer: (C). Row 1 of a is [4 5 6 7]; reversing the columns gives [7 6 5 4], and index 2 is 5.

My answer

Q10. Mid exam (important questions)

If `arr` is a NumPy array, what does `arr[:,-3:]` refer to?

(A) Third element from the last of a one-dimensional array

(B) Third column from the last of a two-dimensional array

(C) Last three elements of a one-dimensional array

(D) Last three columns of a two-dimensional array

Answer: (D). `:` takes all rows and `-3:` takes the columns from the third-last to the end.

My answer

Q11. Mid exam (important questions)

Running `print(X)` displayed the array below, where X is a NumPy array. The bold part is the block `[[5 6 7] [9 10 11]]`.

```
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]
 [13 14 15 16]]
```

What is the CORRECT code to slice the bold part of the array X?

(A) `print(X[1:-1, :-1])`

(B) `print(X[1:-1, :2])`

(C) `print(X[1:3, :2])`

(D) `print(X[-3:-2, -4:-2])`

Answer: (A). Rows 1–2 (`1:-1`) and columns 0–2 (`:-1`). Options B and C stop at column 1, and D returns only [[5 6]].

My answer

Q12.

Mid exam (important questions)

What is the primary requirement for using the `reshape()` method in NumPy?

(A) The new reshaped array must have the same size as the original array.

(B) The new array must be one-dimensional.

(C) The original array must be one-dimensional.

(D) The original array must contain only integers.

Answer: (A). The total number of elements must stay the same (e.g. 12 elements can become 3×4 or 2×6 but not 5×3).

My answer

Q13.

Mid exam (important questions)

What is the output of the following code?

```
import numpy as np
a = np.array(range(12))
b = a.reshape(-1,4)
print(b)
```

(A) `[[ 0 1 2 3] [ 4 5 6 7] [ 8 9 10 11]]`

(B) `[[ 0 1 2] [ 3 4 5] [ 6 7 8] [ 9 10 11]]`

(C) `[[ 1 2 3 4] [ 5 6 7 8] [ 9 10 11 12]]`

(D) `[[ 1 2 3 4 5 6] [ 7 8 9 10 11 12]]`

Answer: (A). The values are 0–11; 4 columns are fixed and `-1` lets NumPy work out $12/4 = 3$ rows.

My answer

Q14.

Sample Mid

What is the output of the following code?

```
import numpy as np
a = np.random.randint(10, size=(4, 3))
b = a.reshape(2,-1)
print(b.shape)
```

(A) `(6, 2)`

(B) `(2, 6)`

(C) `(3, 4)`

(D) `(2, 5)`

Answer: (B). a has 12 elements; with 2 rows, `-1` gives $12/2 = 6$ columns.

My answer

Q15.

Quiz 1 sample (TOPIC 2 & 3)

Each option below starts with `import numpy as np` and `np.random.seed(0)`, and ends with `print(arr1)`. Which of the following codes will produce the array

```
[[4 0 3 3]
 [3 1 3 2]
 [4 0 0 4]]
```

(A) `arr1 = np.random.randint(5,size=(2,4)).reshape(3,4)`

(B) `arr1 = np.random.randint(3,size=(2,6)).reshape(3,4)`

(C) `arr1 = np.random.randint(5,size=(12)).reshape(3,4)`

(D) `arr1 = np.random.randint(3,size=(4,3)).reshape(3,4)`

Answer: (C). The array has values up to 4, so the upper bound must be 5 (B and D only give 0–2), and A fails because 8 elements cannot be reshaped to (3,4).

My answer

Q16.

Topic 2 practice questions

What is the output of the following code?

```
M = np.array([[1, 2, 3],[5, 6, 7]])
a = np.array([[2, -2, -1],[3,4,-3]])
print(f' {M*a}')
```

(A) `[[2 -4 -3] [15 24 -21]]`

(B) `[[3 0 2] [8 10 4]]`

(C) `[[3 8 -9] [15 24 -21]]`

(D) `[[2 -4 -3] [5 -12 -7]]`

Answer: (A). $*$ multiplies element by element: $1 \cdot 2,\ 2 \cdot (-2),\ 3 \cdot (-1)$ and $5 \cdot 3,\ 6 \cdot 4,\ 7 \cdot (-3)$.

My answer

Q17.

Quiz 1 key (T251)

What is the output of the following code?

```
import numpy as np
a = np.array([20, 5, 13, 12])
print( a - (np.sum(a) / np.min(a) - a[1]) )
```

(A) `[30., 15., 23., 22.]`

(B) `[15., 0., 8., 7.]`

(C) `[12., -3., 5., 4.]`

(D) `[5., -10., -2., -3.]`

Answer: (B). sum = 50, min = 5, so $50/5 - 5 = 5.0$, and $a - 5.0$ = [15. 0. 8. 7.].

My answer

Q18.

Sample Mid

Which of the following statements best describes the broadcasting rule in vectorization involving multiple arrays, where the smaller array is 1D?

(A) If two arrays have different dimensions, the smaller dimension array will be expanded with zeros until both arrays have the same dimensions.

(B) In broadcasting, the dimensions of both arrays must be identical, and if they are not, the operation will result in an error.

(C) Broadcasting allows operations between arrays of different shapes by duplicating the smaller array along the dimensions of the larger array.

(D) Broadcasting involves reshaping both arrays to have the same dimensions, regardless of their original shapes.

Answer: (C). The smaller array is (virtually) repeated along the missing or size-1 dimension; it is not padded with zeros.

My answer

Q19.

Quiz 1 key (Version B)

What is the output?

```
import numpy as np
A = np.array([[1, 2, 3],
              [4, 5, 6]])
b = np.array([10, 20, 30])
print(A + b)
```

(A) `[[11 12 13] [24 25 26]]`

(B) `[[10 20 30] [10 20 30]]`

(C) `[[11 22 33] [14 25 36]]`

(D) `[[11 21 31] [14 24 34]]`

Answer: (C). The row vector b is broadcast and added to each row of A.

My answer

Q20.

Quiz-1 (T253, Version A)

What is the output of the following code?

```
A = np.ones((3, 1, 4))
B = np.ones((2, 4))
print(A+B)
```

(A) An array of shape (3, 1, 4)

(B) An array of shape (3, 2, 4)

(C) An array of shape (2, 3, 4)

(D) Error

Answer: (B). Aligning shapes from the right gives (3,1,4) and (1,2,4); each size-1 axis stretches, so the result has shape (3, 2, 4).

My answer

Q21.

Midterm 223

What is the output of the following code?

```
import numpy as np
a = np.array([1, 2, 3, 4])
b = np.array([[6, 4, 7, 9], [13, 5, 6, 8]])
print(b - 2*a + min(a))
```

(A) `[[ 5 1 2 2] [12 2 1 1]]`

(B) `[[11 5 9 11] [25 7 7 9]]`

(C) `[[ 6 1 -1 -6] [13 2 -2 -7]]`

(D) The operation is invalid (results in an error)

Answer: (A). `2*a` = [2 4 6 8] is broadcast to both rows of b, then the scalar `min(a)` = 1 is added.

My answer

Q22.

Mid exam (important questions)

A subset of elements can be selected from a NumPy array based on a given condition by using which of the following methods?

(A) Masking

(B) Slicing

(C) Indexing

(D) Broadcasting

Answer: (A). A Boolean mask such as `arr[arr > 5]` selects the elements that meet a condition (Boolean indexing).

My answer

Q23.

Mid exam (important questions)

What is the output of the following code?

```
import numpy as np
M = np.array([[3, 9, 1, 4], [0, 5, 0, 6], [-9, 4, 8, -6]])
N = abs(M[-1,:])
print(N [N <= 6])
```

(A) `4`

(B) `[3 1 4]`

(C) `[4 6]`

(D) `[False True False True]`

Answer: (C). The last row is [-9 4 8 -6]; its absolute value is [9 4 8 6], and the values ≤ 6 are [4 6].

My answer

Q24.

Sample Mid

What is the output of the following code?

```
import numpy as np
arr1 = np.array([10, 20, 30, 40, 50])
mask = arr1 < 30
result = arr1[mask] * 2
print(result)
```

(A) `[20, 40]`

(B) `[20, 40, 30, 40, 50]`

(C) `[20, 40, 60]`

(D) `[10, 20, 30, 40, 50]`

Answer: (A). Only 10 and 20 are < 30; doubling them gives [20 40].

My answer

Q25.

Topic 2 MCQ (T212)

Given the following array

```
a = np.array([[2, 9, 1, 4],
              [0, 5, 0, 6],
              [1, 4, 9, 5]])
```

what is the output of the code `a[a[:,2] < 5]`?

(A) `[1 0]`

(B) `[2, 1, 4, 0, 0, 1, 4]`

(C) `[[2, 9, 1, 4], [0, 5, 0, 6]]`

(D) `[True, True, False]`

Answer: (C). Column 2 is [1 0 9], so the mask is [True True False], and it selects the first two whole rows.

My answer

Q26.

Midterm 223

Which of the following options can display every other column of `arr` for the 3rd row, 1st row and 4th row (in that order), where

```
arr = np.array([[1, 2, 2], [3, 1, 5], [9, 5, 2], [8, 8, 3]])
```

(A) `arr[[2, 0, 3], ::2]`

(B) `arr[[2, 0, 3], ::-1]`

(C) `arr[::2, [2, 0, 3]]`

(D) `arr[::2, [3, 1, 4]]`

Answer: (A). Fancy indexing with the row list [2, 0, 3] picks the 3rd, 1st and 4th rows, and `::2` takes columns 0 and 2.

My answer

Q27.

Quiz 1 key (Version B)

What is the output?

```
import numpy as np
A = np.arange(10)
B = A[2:7]
B[0] = 99
print(A[2], A[0], B[-1])
```

(A) `2 0 6`

(B) `99 0 6`

(C) `99 0 7`

(D) `99 99 6`

Answer: (B). Slicing creates a view, so changing `B[0]` also changes `A[2]`; `B[-1]` is `A[6]` = 6.

My answer

Q28.

Sample Mid

`.vstack()` can be used for

(A) Row wise joining of two arrays

(B) Column wise joining of two arrays

(C) Breaking an array

(D) Slicing an array

Answer: (A). `vstack` stacks arrays vertically (row-wise); `hstack` joins them column-wise.

My answer

Q29.

Quiz 1 (T251, ISE 291)

Which of the following statements is incorrect?

(A) `.hsplit()` method is used for breaking/splitting the array row wise.

(B) `.concatenate()` function is used to join two or more arrays of the same shape

(C) `.vstack()` can be used for row wise joining

(D) `.hstack()` can be used for column wise joining

Answer: (A). `hsplit` splits horizontally, i.e. into groups of columns; `vsplit` splits into groups of rows.

My answer

Q30. Topic 2 practice questions

What is the output of the following code?

```
x = [1,2,3,5,7,8,12,15]
x1, x2, x3 = np.split(x, [5,7])
print(x1, x2, x3)
```

(A) `[1 2 3 5 7] [8 12] [15]`

(B) `[1 2 3 5] [7 8] [12 15]`

(C) `[1 2 3 4 5] [7] [12 15]`

(D) `[1 2 3 5] [7 8 12] [15]`

Answer: (A). The split points are indices 5 and 7, giving `x[:5]`, `x[5:7]` and `x[7:]`.

My answer

Q31. T253 Sec 03 Quiz 1

Given the below variable definitions, which of the following concatenation commands will successfully execute *without* causing a dimension mismatch error?

```
import numpy as np
a = np.ones((5, 4))
b = np.ones((5, 2))
c = np.ones((2, 5))
```

(A) `np.concatenate((a, b, c), axis=0)`

(B) `np.concatenate((a, b), axis=1)`

(C) `np.concatenate((a, c), axis=0)`

(D) `np.concatenate((b, c), axis=1)`

Answer: (B). Joining along axis 1 only needs the same number of rows: a and b both have 5, giving shape (5, 6).

My answer

Q32. Mid exam (important questions)

What is the purpose of the `np.random.seed(0)` function in the following code?

```
np.random.seed(0)
data = np.random.rand(1000)
plt.hist(data, bins=30, color='blue')
plt.show()
```

(A) It ensures the reproducibility of the random numbers generated.

(B) It forces to get different random numbers every time we execute the code.

(C) It generates a histogram of the data.

(D) It calculates the mean of the random numbers.

Answer: (A). Fixing the seed makes `np.random` produce the same sequence on every run.

My answer

Q33.

Sec 04/05 Quiz 2 (T253)

Which of the following is used to display a plot in Matplotlib?

(A) `plt.display()`

(B) `plt.show()`

(C) `plt.figure()`

(D) `plt.plot()`

Answer: (B). `plt.show()` displays the figure; `plt.figure()` only creates one and `plt.plot()` only draws on it.

My answer

Q34.

Midterm Practice Questions 2

If we want to generate a histogram for the variable `scores` with 10 bins, which code segment should we use?

(A)
```
scores = [88, 92, 79, 85, 94, 76, 81, 89, 93, 90]
plt.figure(); plt.hist(scores, bins=10); plt.show()
```

(B)
```
scores = [88, 92, 79, 85, 94, 76, 81, 89, 93, 90]
plt.figure(scores); plt.show()
```

(C)
```
scores = [88, 92, 79, 85, 94, 76, 81, 89, 93, 90]
plt.figure(); plt.show(scores, bins=10)
```

(D)
```
scores = [88, 92, 79, 85, 94, 76, 81, 89, 93, 90]
plt.hist(scores, range=(1,10)); plt.show()
```

Answer: (A). The data and the number of bins go to `plt.hist`. B and C never call `hist`, and D limits the bins to the range 1–10, which contains none of the scores.

My answer

Q35.

Mid exam (important questions)

In the context of drawing a histogram for scores, which of the following commands creates bins of 41–60, 61–80, 81–100?

```
Score = [48, 62, 79, 65, 81, 94, 56, 91, 69, 99, 60, 84]
```

(A) `plt.hist(Score, bins=[41, 61, 81, 100])`

(B) `plt.hist(Score, bins=4)`

(C) `plt.hist(Score, bins=[41, 61, 81])`

(D) `plt.hist(Score, bins=[40, 60, 80, 100])`

Answer: (A). Bin edges [41,61), [61,81), [81,100] match 41–60, 61–80, 81–100. In D, 60 and 80 fall into the next bin.

My answer

Q36.

Sample Mid

Given the Python code below, how many bins will the histogram display?

```
import matplotlib.pyplot as plt
data = [22, 25, 27, 29, 21, 35, 37, 42, 44, 48]
bins = [20, 30, 40, 50]
plt.hist(data, bins=bins)
```

(A) 2

(B) 4

(C) 3

(D) 5

Answer: (C). 4 edges define 3 bins: [20,30), [30,40), [40,50].

My answer

Q37.

Sample Mid

A histogram can automatically determine the number of bars to show based on the input data. Study the code below and find the number of bars that will be shown on the plot.

```
Labels = ["Q" + str((i % 3) + 1) for i in range(50)]
plt.hist(Labels)
plt.show()
```

(A) 1

(B) 2

(C) 3

(D) 4

Answer: (C). Labels has only three distinct values ('Q1', 'Q2', 'Q3'), so only three bars appear.

My answer

Q38.

Sample Mid

Which parameter in Matplotlib's `plt.pie()` function is used to display the percentage of each slice on the pie chart?

(A) `label`

(B) `autopct`

(C) `format`

(D) `percentage`

Answer: (B). For example, `autopct='%1.1f%%'` writes each slice's percentage with one decimal place.

My answer

Q39.

Quiz 1 Sec 04 solution (T251)

What is the output of the following code?

```
import numpy as np
import matplotlib.pyplot as plt
Subj = ['M', 'P', 'C', 'P', 'P', 'P', 'C', 'P', 'M', 'P', 'M', 'P', 'P', 'M', 'P', 'P', 'C',
    'C', 'M', 'C']
subj_arr = np.array(Subj)
subj_labels, subj_counts = np.unique(Subj, return_counts=True)
plt.figure(figsize=(4,4))
plt.pie(subj_counts, labels = subj_labels, autopct='%.2f%%')
plt.show()
```

(A) A pie chart with 30%, 30% and 40% sectors

(B) A pie chart with 35%, 35% and 30% sectors

(C) A pie chart with 50%, 25% and 25% sectors

(D) A pie chart with 3 equal sectors

Answer: (C). The counts are C = 5, M = 5 and P = 10 out of 20, giving 25%, 25% and 50%.

My answer

Q40.

Quiz 1 Sec 4 (TOPIC 2 & 3)

How do you plot a boxplot using the Matplotlib library?

(A) `plt.boxplot`

(B) `plt.pie`

(C) `plt.hist`

(D) none of the above

Answer: (A). `plt.boxplot(data)` draws a box plot (a pandas DataFrame can also use `df.plot.box()`).

My answer

Part B: Programming

Q41.

Quiz #01 notebook

Create a random integer array in the interval [0, 20] (both ends included) of shape 3×4. Set the random seed to 100. Print the array, its shape, its number of dimensions and its size.

Answer:

```
import numpy as np
np.random.seed(100)
B1 = np.random.randint(0, 21, size=(3, 4))   # high is exclusive, so 21 includes 20
print(B1)
print(B1.shape, B1.ndim, B1.size)
```

My answer

Q42.

HW-01 solutions (T253)

(a) Set the random seed to 1 and create a NumPy array `X` of random integers from 11 to 100 (inclusive) with 7 rows and 9 columns.

(b) From every other row of X, starting at row index 1, select the columns with indices 5, 2 and 7 (in this order).

(c) Select the third column of X, sort it in ascending order, and print it in descending order.

(d) Using a Boolean mask, print all elements of X that are less than or equal to 40.

Answer:

```
import numpy as np
np.random.seed(1)
X = np.random.randint(11, 101, size=(7, 9))
print(X)

# (b) fancy indexing with a row slice and a column list
print(X[1::2, [5, 2, 7]])

# (c)
col = np.sort(X[:, 2])
print(col[::-1])

# (d)
mask = X <= 40
print(X[mask])
```

My answer

Q43.

HW-01 solutions (T253)

Use the array from the previous question:

```
np.random.seed(1)
X = np.random.randint(11, 101, size=(7, 9))
```

(a) For the columns with indices 1, 3 and 6, print the quotients of their entries after integer division by 10.

(b) Split X into three sub-arrays along the rows at row indices 1 and 5 (`axis=0`), then print each part and its shape.

(c) Compute the minimum and the maximum of each column, and append them as two new rows at the bottom of X using `np.concatenate`.

Answer:

```
import numpy as np
np.random.seed(1)
X = np.random.randint(11, 101, size=(7, 9))

# (a)
for i in [1, 3, 6]:
    print(f"Column {i} // 10:", X[:, i] // 10)

# (b)
parts = np.split(X, [1, 5], axis=0)
for p in parts:
    print(p, p.shape)

# (c)
col_min = np.min(X, axis=0)
col_max = np.max(X, axis=0)
X_min_max = np.concatenate((X, [col_min], [col_max]), axis=0)
print(X_min_max)
```

My answer

Q44. Quiz 1 notebook

For a knee surgery, a doctor received the data below (sex, age and severity of each patient). Using NumPy arrays and Boolean masks, print the sex and severity of:

1. patients whose age is between 19 and 22 years (inclusive),
2. patients who are either 18 years old or younger, or older than 22 years,
3. patients older than 20 years.

```
sex=['M','F','M','M','F','M','F','M','M','M','F','M','F','F','M','M','M','F','F','F']
age=[23,19,14,35,21,21,26,23,20,24,21,15,15,15,17,17,23,27,15,18]
Severity=['Low','Low','High','Medium','Low','High','Low','Medium','High','Low','Low','Medium
    ','High','Low,High','Low','Medium','High','Low','Low','Medium']
```

Answer:

```
import numpy as np
npAge, npSex, npSev = np.array(age), np.array(sex), np.array(Severity)

mask = (npAge >= 19) & (npAge <= 22)
print(npSex[mask], npSev[mask])

mask = (npAge <= 18) | (npAge > 22)
print(npSex[mask], npSev[mask])

mask = npAge > 20
print(npSex[mask], npSev[mask])
```

My answer

Q45.

Midterm 213 notebook

Consider the following two arrays:

```
import numpy as np
arr1 = np.array([[3,4,9], [10,9,6]])
arr2 = np.array([[5,4,1], [12,10,11]])
```

a. Create and print a new array `arr3` by stacking arr1 and arr2 column-wise. After that, sort arr3 row-wise.

b. Create and print a new one-dimensional array `arr4` by reshaping arr3.

c. Create and print a new array `arr5` by taking the cube of arr4.

d. Using a mask, print the elements of arr5 that are strictly between 500 and 1000.

Answer:

```
arr3 = np.hstack((arr1, arr2))      # shape (2, 6)
arr3.sort(axis=1)                   # sort each row
print(arr3)

arr4 = arr3.reshape(-1)             # 1-D, 12 elements
print(arr4)

arr5 = arr4 ** 3
print(arr5)

mask = (arr5 > 500) & (arr5 < 1000)
print(arr5[mask])                   # [729 729]
```

My answer

Q46.

Midterm Set-1 notebook

Draw the histogram of `Score_3` using 10 bins in total. Add a title and axis labels.

```
Score_3 = [71, 70, 79, 73, 67, 68, 65, 64, 70, 50, 73, 73, 90, 52, 73, 66, 71., 60, 89, 57,
    78, 63]
```

Answer:

```
import matplotlib.pyplot as plt
plt.figure()
plt.hist(Score_3, bins=10, edgecolor='black')
plt.title('Histogram of Score_3')
plt.xlabel('Score')
plt.ylabel('Frequency')
plt.show()
```

My answer

Q47.

Midterm Set-1 notebook

Draw a pie chart of `Major`. Show the percentage of each category on the plot with one digit after the decimal point, e.g. “12.5%”.

```
Major = ['COE', 'ISE', 'ISE', 'COE', 'CS', 'COE', 'CS', 'ISE', 'CS', 'ISE', 'SWE', 'SWE', '
    COE', 'COE', 'SWE', 'ISE', 'COE', 'CS', 'COE', 'CS', 'ISE', 'CS']
```

Answer:

```
import numpy as np
import matplotlib.pyplot as plt
```

```
labels, counts = np.unique(Major, return_counts=True)
plt.figure()
plt.pie(counts, labels=labels, autopct='%.1f%%')
plt.title('Students by major')
plt.show()
```

My answer

Q48.

Quiz 1 notebook

Three types of sleeping pills (A, B and C) are tested on 12 patients, and the time to sleep is recorded for each pill. Show the box plots of the time to sleep for the three pills in a single figure, label the boxes A, B and C, and add a title.

```
Time_A = [13,12,13,11,12,13,14,14,11,12,15,10]
Time_B = [12,13,16,17,12,14,13,15,16,17,16,17]
Time_C = [20,24,20,22,21,21,25,20,23,25,22,24]
```

Answer:

```
import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(5, 5))
ax.boxplot([Time_A, Time_B, Time_C])
ax.set_xticks([1, 2, 3])
ax.set_xticklabels(['A', 'B', 'C'])
ax.set_title('Sleeping Pills')
ax.set_ylabel('Time to sleep')
plt.show()
```

My answer

Q49.

[new]

Using the object-oriented (OO) approach (`fig, ax = plt.subplots()`):

(a) Create `x` with 100 equally spaced values from 0 to 10, and plot $\sin x$ and $\cos x$ on the same axes with the labels 'sin' and 'cos'.

(b) Add the title 'Sin and Cos', the axis labels 'x' and 'y', a legend in the upper right corner, and a grid.

(c) Add a red dashed vertical reference line at $x = 5$, a green dotted horizontal line at $y = 0$, and the text 'x = 5' near the point (5, 0.5).

Answer:

```
import numpy as np
import matplotlib.pyplot as plt
x = np.linspace(0, 10, 100)
fig, ax = plt.subplots(figsize=(8, 4))
ax.plot(x, np.sin(x), label='sin')
ax.plot(x, np.cos(x), label='cos')
ax.set_title('Sin and Cos')
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.legend(loc='upper right')
ax.grid(True)
ax.axvline(x=5, color='red', linestyle='--')
ax.axhline(y=0, color='green', linestyle=':')
ax.text(5, 0.5, 'x = 5', fontsize=12, color='blue')
plt.show()
```

My answer

Q50. [new]

Consider the system of linear equations

$$2x + y - z = 1, \qquad x + 3y + 2z = 13, \qquad x + z = 4.$$

(a) Write the coefficient matrix `A` and the right-hand side vector `b` as NumPy arrays.

(b) Print the determinant and the inverse of A using `np.linalg`.

(c) Solve the system with `np.linalg.solve` and check the solution by computing `A @ sol`.

(d) Show the difference between `A * A` (element-wise) and `A @ A` (matrix multiplication).

Answer:

```
import numpy as np
A = np.array([[2, 1, -1],
              [1, 3,  2],
              [1, 0,  1]])
b = np.array([1, 13, 4])

print("det:", np.linalg.det(A))          # 10 (up to rounding)
print("inverse:\n", np.linalg.inv(A))

sol = np.linalg.solve(A, b)
print("solution:", sol)                   # [1. 2. 3.]
print("check:", A @ sol, np.allclose(A @ sol, b))

print("element-wise:\n", A * A)
print("matrix product:\n", A @ A)         # same as np.dot(A, A)
```

My answer